

CyberPower Grid-Tied PV Inverter Modbus Protocol

Apply to

CPSPV3000ETLA
CPSPV3600ETLA
CPSPV6000ETLA
CPSPV8000ETLA
CPSPV10000ETLA
CPSPV12000ETLA
CPSPV15000ETLA
CPSPV22000ETLA-WB
CPSPV30000ETLA-WB
CPSPV33000ETLA-WB
CPSPV36000ETLA-WB
CPSPV50000ETL-WB
CPSPV55000ETL-WB
CPSPV66000ETL-WB
CPSPV120KETL
CPSPV140KETL

Revision History

Rev.	Date	Description of Change	Protocol Version	Fault Table Version	Author
V1.00	2018/04/17	First draft	1.00	1.00	Jim.Kuo
V1.01	2018/10/19	Add Fault Table description	1.01	1.00	Jim.Kuo
V1.02	2019/12/11	Add String's E-today & E-total (0x2070~0x209F)	1.02	1.00	Jim.Kuo
V1.03	2024/01/21	Add Load Power Info.	1.03	1.00	Jim.Kuo
V1.04	2024/5/13	1. Add PV5~9 Information(0x4500~0x45FF) 2. Modify Today/Life Energy's unit to 0.1kWh for PV120K & PV140K 3. Modify Event Table	1.04	1.00	Jim.Kuo

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1. Introduction

This document explains the implementation of the Modbus communications protocol specific to CyberPower Systems photovoltaic inverters. The Modbus protocol is an industry standard that allows a master station to interrogate devices for its publically available data and set supported control and configuration parameters.

2. Related documents

<http://www.modbus.org/> Modicon Modbus Protocol Reference Guide

<http://www.modbus.org/> Modbus over Serial Line Specification and Implementation Guide, V1.2

<http://www.modbus.org/> Modbus Application Protocol Specification, V1.1b

<http://www.sunspec.org/> SunSpec Alliance Specifications

3. Modbus Configuration

3.1. Supported Transmission Medium

Products supporting Modbus communicate primarily via the RS-485 (TIA/EIA-485-A) communications standard. The RS-485 medium allows multiple devices to communicate on a single medium using a master/slave approach for arbitrating the bus. All CyberPower Systems products are 1/8 unit load (UL) devices. Depending on the mix of other devices from other vendors, biasing or polarization of devices on the network, this will allow for as few as 32 devices or as many as 255 devices on a single bus.

3.2. Supported Modes of Transmission

The Modbus protocol supports two serial transmission modes, ASCII and RTU (Remote Terminal Unit). CyberPower Systems only supports the RTU mode of operation.

3.3. Communications Parameters

Serial communications default to 8 data bits, no parity, and 1 stop bit. A default baud rate of 9600 BPS is used. Parameters are reconfigurable.

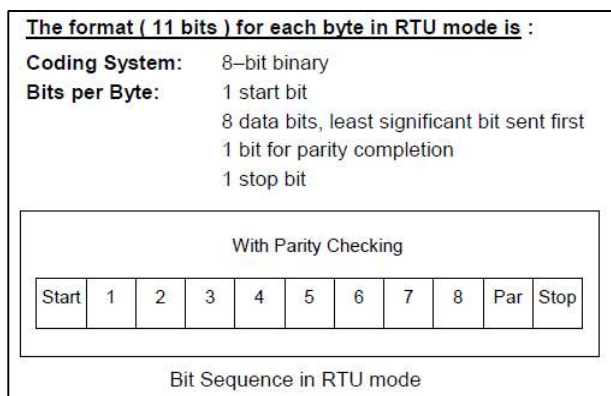


Figure 3-1 The format for each byte in RTU mode

3.4. Exception Responses

Except for broadcast messages, when a master device sends a query to a slave device it expects a normal response. One of four possible events can occur from the master's query:

- If the slave device receives the query without a communication error, and can handle the query normally, it does not return any response.
- If the slave does not receive the query due to a communication error, no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query, but detects a communication error (parity, LRC, or CRC), no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query without a communication error, but cannot handle it (for example, if the

request is to read a non-existent coil or register), the slave will return an exception response informing the master of the nature of the error.

The table below describes the exception codes supported by CyberPower Systems devices along with their possible causes.

Code	Name	Meaning
01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action for the slave. If a Poll Program Complete command was issued, this code indicates that no program function preceded it.
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the slave.
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for the slave.
04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the slave was attempting to perform the requested action.
06	SLAVE DEVICE BUSY	The slave is engaged in processing a long-duration program command. The master should retransmit the message later when the slave is free.

Table 3-1 Exception Codes Supported by CyberPower Systems Devices

3.5. CRC Generation

When RTU mode is used for character framing, the error checking field contains a 16-bit value implemented as two 8-bit bytes. The error check value is the result of a Cyclical Redundancy Check calculation performed on the message contents.

The CRC field is appended to the message as the last field in the message. When this is done, the low-order byte of the field is appended first, followed by the high-order byte. The CRC high-order byte is the last byte to be sent in the message.

The CRC value is calculated by the transmitting device, which appends the CRC to the message. The receiving device recalculates a CRC during receipt of the message, and compares the calculated value to the actual value it received in the CRC field. If the two values are not equal, an error results.

3.6. Addressing Rules

The Modbus addressing space comprises 256 different addresses. The addresses from 248 to 255 are reserved and the address 0 is broadcast address.

All CyberPower Systems products support broadcast request packets from the master. As its name implies, broadcasts allow all devices to receive and process the same command from the Modbus master. Broadcasts are triggered by setting the slave address to 0 (0x00).

0	From 1 to 247	From 248 to 255
Broadcast address	Slave individual addresses	Reserved

Table 3-2 Addressing rules

3.7. RTU Framing

In RTU mode, messages start with a silent interval of at least 3.5 character times. This is most easily implemented as a multiple of character times at the baud rate that is being used on the network. The first field then transmitted is the device address.

The allowable characters transmitted for all fields are hexadecimal 0–9, A–F. Networked devices monitor the network bus continuously, including during the ‘silent’ intervals. When the first field (the address field) is received, each device decodes it to find out if it is the addressed device. Following the last transmitted character, a similar interval of at least 3.5 character times marks the end of the message. A new message can begin after this interval.

The entire message frame must be transmitted as a continuous stream. If a silent interval of more than 1.5 character times occurs before completion of the frame, the receiving device flushes the incomplete message and assumes that the next byte will be the address field of a new message.

Similarly, if a new message begins earlier than 3.5 character times following a previous message, the receiving device will consider it a continuation of the previous message. This will set an error, as the value in the final CRC field will not be valid for the combined messages.

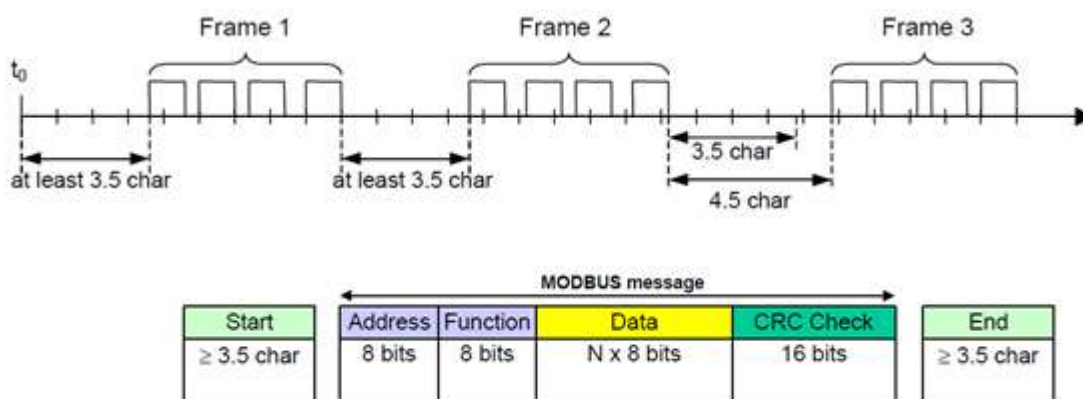


Figure 3-2 RTU Message Frame (1)

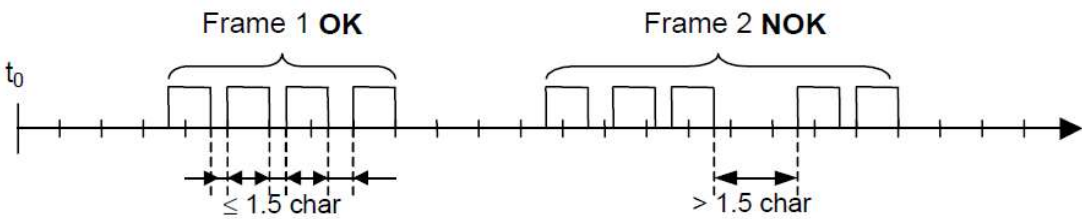


Figure 3-3 RTU Message Frame (2)

4. Supported Modbus Functions

4.1. Function 1 (0x01): Read Coils

Description

Reads the ON/OFF status of discrete outputs.

Query

The query message specifies the starting coil and quantity of coils to be read.

Response

The coil status in the response message is packed as one coil per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the coil addressed in the query. The other coils follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes.

If the returned coil quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Coils 6000h – 600Fh:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	01	Function	01
Starting Address Hi	60	Byte Count	02
Starting Address Lo	00	Outputs status 6000-6007	00
Quantity of Outputs Hi	00	Outputs status 6008-600F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	23	CRC Hi	FC
CRC Hi	C6		

Table 4-1 example of a request to read Coils

4.2. Function 2 (0x02): Read Discrete Inputs

Description

Reads the ON/OFF status of discrete inputs (1X references) in the slave.

Query

The query message specifies the starting input and quantity of inputs to be read.

Response

The input status in the response message is packed as one input per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the input addressed in the query. The other inputs follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes.

If the returned input quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Discrete Inputs 5000h – 500Fh:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	02	Function	02
Starting Address Hi	50	Byte Count	02
Starting Address Lo	00	Outputs status 5000-5007	00
Quantity of Outputs Hi	00	Outputs status 5008-500F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	68	CRC Hi	B8
CRC Hi	C6		

Table 4-2 example of a request to read Discrete Inputs

4.3. Function 3 (0x03): Read Holding Registers

Description

Reads the binary contents of holding registers (4X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Holding Registers 0532h – 0533h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>

Slave ID	01	Slave ID	01
Function	03	Function	03
Starting Address Hi	05	Byte Count	04
Starting Address Lo	32	Register value Hi (0532)	07
No. of Registers Hi	00	Register value Lo (0532)	D0
No. of Registers Lo	02	Register value Hi (0533)	00
CRC Lo	65	Register value Lo (0533)	01
CRC Hi	08	CRC Lo	3B
		CRC Hi	7E

Table 4-3 example of a request to read Holding Registers

4.4. Function 4 (0x04): Read Input Registers

Description

Reads the binary contents of input registers (3X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Input Registers 1000h – 1001h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	04	Function	04
Starting Address Hi	10	Byte Count	04
Starting Address Lo	00	Register value Hi (0532)	CC
No. of Registers Hi	00	Register value Lo (0532)	94
No. of Registers Lo	02	Register value Hi (0533)	06
CRC Lo	75	Register value Lo (0533)	0C
CRC Hi	0B	CRC Lo	86
		CRC Hi	9D

Table 4-4 example of a request to read Input Registers

4.5. Function 5 (0x05): Write Single Coil

Description

Forces a single coil to either ON or OFF. When broadcast, the function forces the same coil reference in all attached slaves.

Query

The query message specifies the coil reference to be forced. The requested ON/OFF state is specified by a constant in the query data field. A value of FF 00 hex requests the coil to be ON. A value of 00 00 requests it to be OFF. All other values are illegal and will not affect the coil.

Response

The normal response is an echo of the query, returned after the coil state has been forced. Here is an example of a request to write Coil 6000h ON:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	05	Function	05
Output Address Hi	60	Output Address Hi	60
Output Address Lo	00	Output Address Lo	00
Output Value Hi	FF	Output Value Hi	FF
Output Value Lo	00	Output Value Lo	00
CRC Lo	92	CRC Lo	92
CRC Hi	3A	CRC Hi	3A

Table 4-5 example of a request to write single Coil

4.6. Function 6 (0x06): Write Single Register

Description

This function code is used to write a single holding register in a remote device. When broadcast, the function presets the same register reference in all attached slaves.

Query

The query message specifies the register reference to be preset. The requested preset value is specified in the query data field.

Response

The normal response is an echo of the request, returned after the register contents have been written.

Here is an example of a request to write Holding Register 0512h to 03E8h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	06	Function	06
Register Address Hi	05	Register Address Hi	05
Register Address Lo	12	Register Address Lo	12
Register Value Hi	03	Register Value Hi	03
Register Value Lo	E8	Register Value Lo	E8
CRC Lo	29	CRC Lo	29
CRC Hi	BD	CRC Hi	BD

Table 4-6 example of a request to write Holding Register

4.7. Function 16 (0x10): Write Multiple Registers

Description

This function code is used to write a block of contiguous registers (1 to 123 registers). When broadcast, the function presets the same register references in all attached slaves.

Query

The requested written values are specified in the request data field. Data is packed as two bytes per register.

Response

The normal response returns the function code, starting address, and quantity of registers written.

Here is an example of a request to write three registers at 0532h to 07DCh, 0008h and 0014h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	10	Function	10
Starting Address Hi	05	Starting Address Hi	05
Starting Address Lo	32	Starting Address Lo	32
Quantity of Registers Hi	00	Quantity of Registers Hi	00
Quantity of Registers Lo	03	Quantity of Registers Lo	03
Byte Count	06	CRC Lo	21
Registers Value Hi	07	CRC Hi	0B
Registers Value Lo	DC		
Registers Value Hi	00		
Registers Value Lo	08		

Registers Value Hi	00		
Registers Value Lo	14		
CRC Lo	07		
CRC Hi	D0		

Table 4-7 example of a request to write three registers

5. Modbus Register Map

5.1. Modbus Data Formats

The MODBUS specification is not explicit on how to encode numbers other than 16 bit integers. Differences do exist between one manufacturer's implementation and another's. **This specification restricts values to the following standard data formats and it is difference to the formats in "SunSpec MODBUS Register Mappings".**

16 bit Integer Values

Values are stored in big-endian order per the MODBUS specification and consist of a single register. All integer values are documented a signed or unsigned. All signed values are represented using two's-compliment.

Modbus Register	1															
Byte	0								1							
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

32 bit Integer Values

32 bit integers are stored using two registers in big-endian order

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

64 bit Integer Values

64 bit integers are stored using four registers in big-endian order.

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

Modbus Register	3								4							
Byte	4				5				6				7			
Bits	47 ... 40				39 ... 32				63 ... 56				55 ... 48			

String Values

Store variable length string values in a fixed size register range using a NULL (0 value) to terminate or pad the string. For example, up to 16 characters can be stored in 8 contiguous

registers as follows

EX : "EXAMPLE STRING!"

Modbus Register	1		2		3		4		5		6		7		8	
Byte	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Character	E	X	A	M	P	L	E		S	T	R	I	N	G	!	NULL

NOT_IMPLEMENTED value: all registers filled with NULL or 0x0000.

Floating Point Values,

Floating point values are 32 bits and encoded according to the IEEE 754 floating point standard.

Modbus Register	1		2		
Byte	0	1	2	3	
Bits	15 ... 8	7 ... 0	31	30 ... 24	23 ... 16
IEEE 754	Fraction least		sign	Exponent	Fraction

5.2. Input Registers (0x04)

5.2.1. Monitoring Parameters

Address (Hex)	Name	Type	Unit	Remark
114D	Load Power	Int16	1W	Only Support PV3K/3K6/6K
11CA ~11CB	Load Power (Phase A)	int32	1W	Only Support PV8K/10K/12K/15K
11CC ~11CD	Load Power (Phase B)	int32	1W	Only Support PV8K/10K/12K/15K
11CE ~11CF	Load Power (Phase C)	int32	1W	Only Support PV8K/10K/12K/15K

Address (Hex)	Name	Type	Unit	Remark
2100 ~2137	Reserved			
2138 2139	Today Energy (feed in Grid)	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
213A 213B	Today Runtime Reserved	uint32	1sec.	
213C 213D	System Life Energy (feed in Grid)	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
213E 213F	System Life Runtime	uint32	1sec.	
2140 2141	PV1 today energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
2142 2143	PV1 life energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
2144 2145	PV2 today energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
2146 2147	PV2 life energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
2148 2149	PV3 today energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
214A	PV3 life energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is

214B				1Wh
214C	PV4 today energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
214D				
214E	PV4 life energy	uint32	1Wh/0.1kWh	Unit is 0.1kWh in PV120/140k, Other model is 1Wh
214F				
2150 ~2193	Reserved			
2194	PV1 Voltage	uint16	0.1V	
2195	PV1 Current	uint16	0.01A	
2196	PV1 Power	uint16	1W	
2197	PV2 Voltage	uint16	0.1V	
2198	PV2 Current	uint16	0.01A	
2199	PV2 Power	uint16	1W	
219A	PV3 Voltage	uint16	0.1V	
219B	PV3 Current	uint16	0.01A	
219C	PV3 Power	uint16	1W	
219D	PV4 Voltage	uint16	0.1V	
219E	PV4 Current	uint16	0.01A	
219F	PV4 Power	uint16	1W	
21A0	Grid phase-A voltage	uint16	0.1V	
21A1	Grid phase-B voltage	uint16	0.1V	
21A2	Grid phase-C voltage	uint16	0.1V	
21A3	Grid phase-A output current	uint16	0.01A	
21A4	Grid phase-B output current	uint16	0.01A	
21A5	Grid phase-C output current	uint16	0.01A	
21A6	Grid phase-A output power	uint16	1W	
21A7	Grid phase-B output power	uint16	1W	
21A8	Grid phase-C output power	uint16	1W	
21A9	Grid phase-AB voltage	uint16	0.1V	

21AA	Grid phase-BC voltage	uint16	0.1V	
21AB	Grid phase-CA voltage	uint16	0.1V	
21AC	Grid Frequency	uint16	0.01Hz	
21AD	Grid output active power	uint16	1W	Power = value * Power scale factor (address 21B0h)
21AE	Grid output apparent power	uint16	1VA	Power = value * Power scale factor (address 21B0h)
21AF	Grid input reactive power	int16	1VA _r	Power = value * Power scale factor (address 21B0h) '+' means inductive; - means capacitive. From Grid point of view, inverter is a load to Grid
21B0	Power scale factor	uint16		Range: 1-1000
21B1 ~21B8	Reserved			
21B9	BUS1_Voltage	uint16	0.1V	
21BA	BUS2_Voltage	uint16	0.1V	
21BB	PV Rleakage	uint16	1kohm	
21BC	Temperature (air inside around power stage)	int16	0.1°C	
21BD	Temperature (air inside of wiring box)	int16	0.1°C	
21BE	Highest inverter temperature	int16	0.1°C	
21BF	Highest D/D temperature	int16	0.1°C	
21C0	Reserved			
21C1	Reconnection Time	uint16	1sec	
21C2	I_Event_flags1	uint32	bitfield	Event flags for Sunspec protocol
21C3				
21C4	I_Event_flags2	uint32	bitfield	Not supported
21C5				
21C6	I_Event_1_Vend or	uint16	bitfield	Event Flags (E00 ~ E15)

21C7	I_Event_2_Vend or	uint16	bitfield	Event Flags (E16 ~ E31)
21C8	I_Event_3_Vend or	uint16	bitfield	Event Flags (W00 ~ W15)
21C9	I_Event_4_Vend or	uint16	bitfield	Event Flags (F00 ~ F15)
21CA	I_Event_5_Vend or	uint16	bitfield	Event Flags (F16 ~ F31)
21CB	I_Event_6_Vend or	uint16	bitfield	Event Flags (F32 ~ F47)
21CC	I_Event_7_Vend or	uint16	bitfield	Event Flags (F48 ~ F63)
21CD	I_Event_8_Vend or	uint16	bitfield	Event Flags (F64 ~ F79)
21CE	I_Event_9_Vend or	uint16	bitfield	Event Flags (W16 ~ W31)
21CF ~21DF	Reserved			
21E0	I_Event_1_Vend or	uint16	bitfield	Mapped from 21C6but keep flags for 300s
21E1	I_Event_2_Vend or	uint16	bitfield	Mapped from 21C7 but keep flags for 300s
21E2	I_Event_3_Vend or	uint16	bitfield	Mapped from 21C8 but keep flags for 300s
21E3	I_Event_4_Vend or	uint16	bitfield	Mapped from 21C9 but keep flags for 300s
21E4	I_Event_5_Vend or	uint16	bitfield	Mapped from 21CA but keep flags for 300s
21E5	I_Event_6_Vend or	uint16	bitfield	Mapped from 21CB but keep flags for 300s
21E6	I_Event_7_Vend or	uint16	bitfield	Mapped from 21CC but keep flags for 300s
21E7	I_Event_8_Vend or	uint16	bitfield	Mapped from 21CD but keep flags for 300s
21E8	I_Event_9_Vend or	uint16	bitfield	Mapped from 21CE but keep flags for 300s
21E9 ~21FF	Reserved			

Address (Hex)	Name	Type	Unit	Remark
4500	PV5 today energy	uint32	0.1kWh	
4501				
4502	PV5 life energy	uint32	0.1kWh	
4503				
4504	PV6 today energy	uint32	0.1kWh	
4505				
4506	PV6 life energy	uint32	0.1kWh	
4507				
4508	PV7 today energy	uint32	0.1kWh	
4509				
450A	PV7 life energy	uint32	0.1kWh	
450B				
450C	PV8 today energy	uint32	0.1kWh	
450D				
450E	PV8 life energy	uint32	0.1kWh	
450F				
4510	PV9 today energy	uint32	0.1kWh	
4511				
4512	PV9 life energy	uint32	0.1kWh	
4513				
4514	PV10 today energy	uint32	0.1kWh	
4515				
4516	PV10 life energy	uint32	0.1kWh	
4517				
4518	PV11 today energy	uint32	0.1kWh	
4519				
451A	PV11 life energy	uint32	0.1kWh	
451B				
451C	PV12 today energy	uint32	0.1kWh	
451D				
451E	PV12 life energy	uint32	0.1kWh	
451F				
4520 ~454F	Reserved			
4550	PV5 Voltage	uint16	0.1V	
4551	PV5 Current	uint16	0.01A	
4552	PV5 Power	uint16	1W	

4553	PV6 Voltage	uint16	0.1V	
4554	PV6 Current	uint16	0.01A	
4555	PV6 Power	uint16	1W	
4556	PV7 Voltage	uint16	0.1V	
4557	PV7 Current	uint16	0.01A	
4558	PV7 Power	uint16	1W	
4559	PV8 Voltage	uint16	0.1V	
455A	PV8 Current	uint16	0.01A	
455B	PV8 Power	uint16	1W	
455C	PV9 Voltage	uint16	0.1V	
455D	PV9 Current	uint16	0.01A	
455E	PV9 Power	uint16	1W	
455F	PV10 Voltage	uint16	0.1V	
4560	PV10 Current	uint16	0.01A	
4561	PV10 Power	uint16	1W	
4562	PV11 Voltage	uint16	0.1V	
4563	PV11 Current	uint16	0.01A	
4564	PV11 Power	uint16	1W	
4565	PV12 Voltage	uint16	0.1V	
4566	PV12 Current	uint16	0.01A	
4567	PV12 Power	uint16	1W	

Note: Only PV120K & PV140K support 0x4500~0x4567

5.2.2.String Information

Address (Hex)	Name	Type	Unit	Remark
2000	Reserved			
2001	String1-16 connection setting	uint16	Bitfield	bit0: string1 connection bit15: string16 connection 0: not available 1: Connect
2002	String17-32 connection setting	uint16	Bitfield	bit0: string17 connection bit15: string32 connection 0: not available 1: Connect

2003	Reserved			
2004	Reserved			
2005	Reserved			
2006	Reserved			
2007	String1-16 Positive fuse flags	unit16	Bitfield	bit0: string(16n+1) fuse bit15: string(16n+16) fuse 0: normal 1: broken
2008	String1-16 Negative fuse flags	unit16	Bitfield	bit0: string(16n+1) fuse bit15: string(16n+16) fuse 0: normal 1: broken
2009	String17-32 Positive fuse flags	unit16	Bitfield	bit0: string(16n+1) fuse bit15: string(16n+16) fuse 0: normal 1: broken
200A	String17-32 Negative fuse flags	unit16	Bitfield	bit0: string(16n+1) fuse bit15: string(16n+16) fuse 0: normal 1: broken
200B	Reserved			
200C	Reserved			
200D	Reserved			
200E	Reserved			
200F	Reserved			
2010	String 1 voltage	uint16	0.1V	
2011	String 1 current	uint16	0.01A	
2012	String 1 power	uint16	1W	
2013	String 2 voltage	uint16	0.1V	
2014	String 2 current	uint16	0.01A	
2015	String 2 power	uint16	1W	
2016	String 3 voltage	uint16	0.1V	
2017	String 3 current	uint16	0.01A	
2018	String 3 power	uint16	1W	
2019	String 4 voltage	uint16	0.1V	

201A	String 4 current	uint16	0.01A	
201B	String 4 power	uint16	1W	
201C	String 5 voltage	uint16	0.1V	
201D	String 5 current	uint16	0.01A	
201E	String 5 power	uint16	1W	
201F	String 6 voltage	uint16	0.1V	
2020	String 6 current	uint16	0.01A	
2021	String 6 power	uint16	1W	
2022	String 7 voltage	uint16	0.1V	
2023	String 7 current	uint16	0.01A	
2024	String 7 power	uint16	1W	
2025	String 8 voltage	uint16	0.1V	
2026	String 8 current	uint16	0.01A	
2027	String 8 power	uint16	1W	
2028	String 9 voltage	uint16	0.1V	
2029	String 9 current	uint16	0.01A	
202A	String 9 power	uint16	1W	
202B	String 10 voltage	uint16	0.1V	
202C	String 10 current	uint16	0.01A	
202D	String 10 power	uint16	1W	
202E	String 11 voltage	uint16	0.1V	
202F	String 11 current	uint16	0.01A	
2030	String 11 power	uint16	1W	
2031	String 12 voltage	uint16	0.1V	
2032	String 12 current	uint16	0.01A	
2033	String 12 power	uint16	1W	
2034	String 13 voltage	uint16	0.1V	
2035	String 13 current	uint16	0.01A	
2036	String 13 power	uint16	1W	
2037	String 14 voltage	uint16	0.1V	
2038	String 14 current	uint16	0.01A	
2039	String 14 power	uint16	1W	
203A	String 15 voltage	uint16	0.1V	
203B	String 15 current	uint16	0.01A	
203C	String 15 power	uint16	1W	
203D	String 16 voltage	uint16	0.1V	
203E	String 16 current	uint16	0.01A	
203F	String 16 power	uint16	1W	
2040	String 17 voltage	uint16	0.1V	

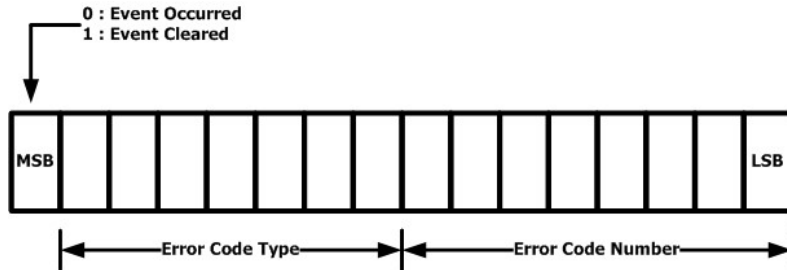
2041	String 17 current	uint16	0.01A	
2042	String 17 power	uint16	1W	
2043	String 18 voltage	uint16	0.1V	
2044	String 18 current	uint16	0.01A	
2045	String 18 power	uint16	1W	
2046	String 19 voltage	uint16	0.1V	
2047	String 19 current	uint16	0.01A	
2048	String 19 power	uint16	1W	
2049	String 20 voltage	uint16	0.1V	
204A	String 20 current	uint16	0.01A	
204B	String 20 power	uint16	1W	
204C	String 21 voltage	uint16	0.1V	
204D	String 21 current	uint16	0.01A	
204E	String 21 power	uint16	1W	
204F	String 22 voltage	uint16	0.1V	
2050	String 22 current	uint16	0.01A	
2051	String 22 power	uint16	1W	
2052	String 23 voltage	uint16	0.1V	
2053	String 23 current	uint16	0.01A	
2054	String 23 power	uint16	1W	
2055	String 24 voltage	uint16	0.1V	
2056	String 24 current	uint16	0.01A	
2057	String 24 power	uint16	1W	
2058	String 25 voltage	uint16	0.1V	
2059	String 25 current	uint16	0.01A	
205A	String 25 power	uint16	1W	
205B	String 26 voltage	uint16	0.1V	
205C	String 26 current	uint16	0.01A	
205D	String 26 power	uint16	1W	
205E	String 27 voltage	uint16	0.1V	
205F	String 27 current	uint16	0.01A	
2060	String 27 power	uint16	1W	
2061	String 28 voltage	uint16	0.1V	
2062	String 28 current	uint16	0.01A	
2063	String 28 power	uint16	1W	
2064	String 29 voltage	uint16	0.1V	
2065	String 29 current	uint16	0.01A	
2066	String 29 power	uint16	1W	
2067	String 30 voltage	uint16	0.1V	

2068	String 30 current	uint16	0.01A	
2069	String 30 power	uint16	1W	
206A	String 31 voltage	uint16	0.1V	
206B	String 31 current	uint16	0.01A	
206C	String 31 power	uint16	1W	
206D	String 32 voltage	uint16	0.1V	
206E	String 32 current	uint16	0.01A	
206F	String 32 power	uint16	1W	
2070 ~2071	String 1 Today Energy	uint32	1Wh	
2072 ~2073	String 1 Life Energy	uint32	1Wh	
2074 ~2075	String 2 Today Energy	uint32	1Wh	
2076 ~2077	String 2 Life Energy	uint32	1Wh	
2078 ~2079	String 3 Today Energy	uint32	1Wh	
207A ~207B	String 3 Life Energy	uint32	1Wh	
207C ~207D	String 4 Today Energy	uint32	1Wh	
207E ~207F	String 4 Life Energy	uint32	1Wh	
2080 ~2081	String 5 Today Energy	uint32	1Wh	
2082 ~2083	String 5 Life Energy	uint32	1Wh	
2084 ~2085	String 6 Today Energy	uint32	1Wh	
2086 ~2087	String 6 Life Energy	uint32	1Wh	
2088 ~2089	String 7 Today Energy	uint32	1Wh	
208A ~208B	String 7 Life Energy	uint32	1Wh	
208C ~208D	String 8 Today Energy	uint32	1Wh	
208E	String 8 Life Energy	uint32	1Wh	

~208F				
2090 ~2091	String 9 Today Energy	uint32	1Wh	
2092 ~2093	String 9 Life Energy	uint32	1Wh	
2094 ~2095	String 10 Today Energy	uint32	1Wh	
2096 ~2097	String 10 Life Energy	uint32	1Wh	
2098 ~2099	String 11 Today Energy	uint32	1Wh	
209A ~209B	String 11 Life Energy	uint32	1Wh	
209C ~209D	String 12 Today Energy	uint32	1Wh	
209E ~209F	String 12 Life Energy	uint32	1Wh	

Appendix A: Error Code

1. Error Code Format



Error Code Type	
E code: Error	45h
W code : Warning	57h
F code : Fault	46h

For example:

- "AC Over Frequency" exist: E07 => 4507h

2. "E" Error Code Table

Error Code	Hex Code	Event	Remark
E00	4500	No Grid	No Utility
E01	4501	Grid over frequency	AC Frequency above limit
E02	4502	Grid under frequency	AC Frequency under limit
E03	4503	Phase-A over voltage	AC Phase A Voltage above limit
E04	4504	Phase-A under voltage	AC Phase A Voltage under limit
E05	4505	Reserved	
E06	4506	Reserved	
E07	4507	Reserved	
E08	4508	Reserved	
E09	4509	Reserved	
E10	450A	Reserved	
E11	450B	Reserved	
E12	450C	Reserved	
E13	450D	Phase-B over voltage	AC Phase B Voltage above limit
E14	450E	Phase-B under voltage	AC Phase B Voltage under limit
E15	450F	Reserved	
E16	4510	Reserved	

E17	4511	Phase-C over voltage	AC Phase C Voltage above limit
E18	4512	Phase-C under voltage	AC Phase C Voltage under limit
E19	4513	Reserved	
E20	4514	PV1 over voltage	Input1 DC Voltage above limit
E21	4515	PV2 over voltage	Input2 DC Voltage above limit
E22	4516	PV3 over voltage	Input3 DC Voltage above limit
E23	4517	PV4 over voltage	Input4 DC Voltage above limit
E24	4518	PV5 over voltage	Input5 DC Voltage above limit
E25	4519	PV6 over voltage	Input6 DC Voltage above limit
E26	451A	PV7 over voltage	Input7 DC Voltage above limit
E27	451B	PV8 over voltage	Input8 DC Voltage above limit
E28	451C	PV9 over voltage	Input9 DC Voltage above limit
E29	451D	Reserved	
E30	451E	Reserved	
E31	451F	Reserved	

3. "W" Error Code Table

Error Code	Hex Code	Event	Remark
W00	5700	PV1 under voltage	Input1 DC Voltage under limit
W01	5701	PV2 under voltage	Input2 DC Voltage under limit
W02	5702	PV3 under voltage	Input3 DC Voltage under limit
W03	5703	PV4 under voltage	Input4 DC Voltage under limit
W04	5704	PV5 under voltage	Input5 DC Voltage under limit
W05	5705	PV6 under voltage	Input6 DC Voltage under limit
W06	5706	PV7 under voltage	Input7 DC Voltage under limit
W07	5707	PV8 under voltage	Input8 DC Voltage under limit
W08	5708	Power de-rating	Reduce output power due to temperature high or else
W09	5709	String fuse break	String fuse break
W10	570A	HW Fan	Fan Failed
W11	570B	SPD break	SPD break
W12	570C	Str PW abn.	String power abnormal
W13	570D	Reserved	
W14	570E	Reserved	
W15	570F	Reserved	

W16	5710	Reserved	
W17	5711	Reserved	
W18	5712	Reserved	
W19	5713	Reserved	
W20	5714	Reserved	
W21	5715	Reserved	
W22	5716	Reserved	
W23	5717	Reserved	
W24	5718	EEPROM Fail	EEPROM read/write fail
W25	5719	Reserved	
W26	571A	Reserved	
W27	571B	Reserved	
W28	571C	PV9 under voltage	Input9 DC Voltage under limit
W29	571D	VPC Derating	Reduce output power due to VPC
W30	571E	AC SPD Break	
W31	571F	Unbalance Charge Bus	

4. "F" Error Code Table

Error Code	Hex Code	Event	Remark
F00	4600	Reserved	
F01	4601	Grid output DC over current	AC Current DC Component above limit
F02	4602	Reserved	
F03	4603	Reserved	
F04	4604	Reserved	
F05	4605	Temperature low	Temperature under limit
F06	4606	Temperature high	Temperature above limit
F07	4607	Temperature sensor fail	Temperature sensor error, or heating circuit failure
F08	4608	Reserved	
F09	4609	Reserved	
F10	460A	Reserved	
F11	460B	Reserved	
F12	460C	Reserved	
F13	460D	Reserved	
F14	460E	Firmwares are not compatible	The firmware are not compatible
F15	460F	DSP ADC1	DSP ADC1 sensor error or circuit failure

F16	4610	DSP ADC2	DSP ADC2 sensor error or circuit failure
F17	4611	DSP ADC3	DSP ADC3 sensor error or circuit failure
F18	4612	DSP ADC4	DSP ADC4 sensor error or circuit failure
F19	4613	Red. ADC1	Redundant MCU ADC1 sensor error or circuit failure
F20	4614	Efficiency	HW Efficiency fail
F21	4615	Reserved	
F22	4616	HW Com1	The serial communication port between DSP and comm. MCU is failure
F23	4617	HW Com2	The serial communication port between DSP and redundant MCU is failure
F24	4618	Ground current	Residual current is above limit
F25	4619	RCMU fail	Residual current sensor error or circuit failure
F26	461A	Insulation fail	Insulation resistance to Earth is under limit
F27	461B	Reserved	
F28	461C	Relay short	Relay or drive circuit failure (Relay Short)
F29	461D	Relay open	Relay or drive circuit failure (Relay Open)
F30	461E	Bus unbalanced	Bus1 and Bus2 voltage are not balance
F31	461F	Bus+ OVR	Bus1 voltage above limit or circuit failure
F32	4620	Bus+ UVR	Bus1 voltage under limit or circuit failure
F33	4621	Bus- OVR	Bus2 voltage above limit or circuit failure
F34	4622	Bus- UVR	Bus2 voltage under limit or circuit failure
F35	4623	Total Bus OVR	Bus voltage above limit or circuit failure
F36	4624	Reserved	
F37	4625	AC over current	AC Current above limit
F38	4626	Reserved	
F39	4627	Reserved	
F40	4628	Reserved	
F41	4629	Reserved	
F42	462A	CT fail	AC Current sensor or circuit failure
F43	462B	Reserved	
F44	462C	Reserved	
F45	462D	AC HW OCP	Hardware AC over current protection
F46	462E	Aux PWR Fail	Aux PWR Fail
F47	462F	HW D/D Fail	HW D/D Fail
F48	4630	Manual shutdown	Manual shutdown
F49	4631	Reserved	
F50	4632	Reserved	
F51	4633	Reserved	

F52	4634	Reserved	
F53	4635	Reserved	
F54	4636	Reserved	
F55	4637	Reserved	
F56	4638	Reserved	
F57	4639	Reserved	
F58	463A	Reserved	
F59	463B	Reserved	
F60	463C	PV Input over current	Input Current above limit
F61	463D	Reserved	
F62	463E	Reserved	
F63	463F	Reserved	
F64	4640	Reserved	
F65	4641	Reserved	
F66	4642	Reserved	
F67	4643	Reserved	
F68	4644	Reserved	
F69	4645	Other	Not defined fault
F70	4646	Reserved	
F71	4647	Reserved	